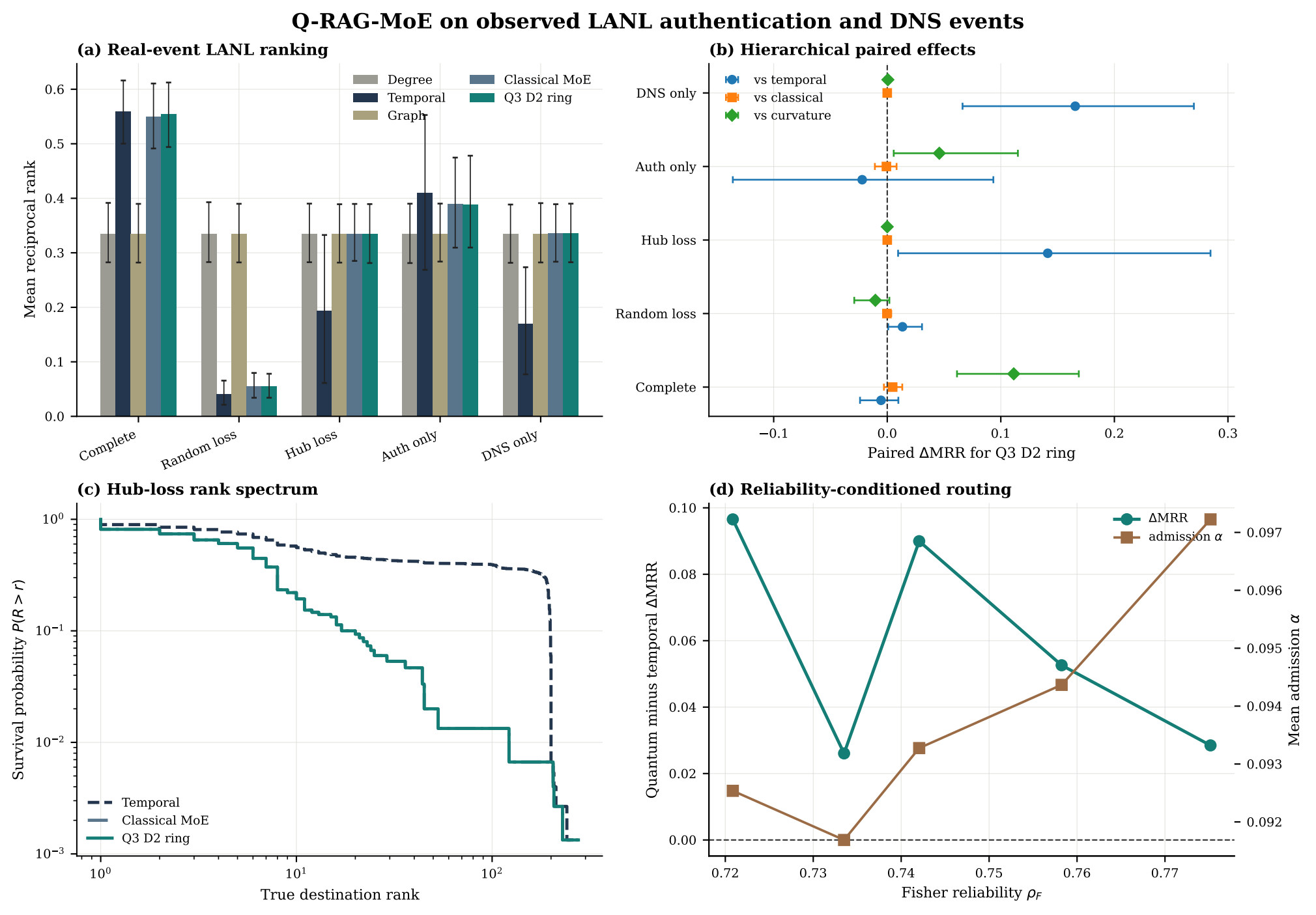
Q-RAG-MoE on observed LANL authentication and DNS events
(a) Real-event LANL ranking
Mean reciprocal rank
Degree
Temporal
Graph
Classical MoE
Q3 D2 ring
Complete
Random loss
Hub loss
Auth only
DNS only
(b) Hierarchical paired effects
vs temporal
vs classical
vs curvature
Paired ΔMRR for Q3 D2 ring
(c) Hub-loss rank spectrum
Survival probability P(R > r)
True destination rank
Temporal
Classical MoE
Q3 D2 ring
(d) Reliability-conditioned routing
Quantum minus temporal ΔMRR
Mean admission α
ΔMRR
admission α
Fisher reliability ρF